Selecting structurally divergent clinical *E. coli* strains for cryo-ET by gene content

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1 Summary

We analysed a set of 2894 *E. coli* genomes, which originated from clinical isolates collected between 2013 and 2018 and in 2020 during an observational study in Oxfordshire (Lipworth et al. 2025).

We selected 2853 genomes based on a set of quality metrics including genome length, quality and taxonomic classification. We created a pangenome, and analysed the differences between them using Jaccard distances on a gene presence/absence matrix. We selected genes based on their functional annotated categories assigned by eggNOG mapper. We chose to include genes in 4 functional categories: cell motility (e.g. flagella, chemotaxis), extracellular structures (e.g. pili), intracellular trafficking, secretion and vesicular transport, and cell wall, membrane and envelope biogenesis.

We chose 3 genomes, which maximise the minimum difference between each of the genomes based on their pairwise Jaccard distances, in addition to *E. coli* K-12.

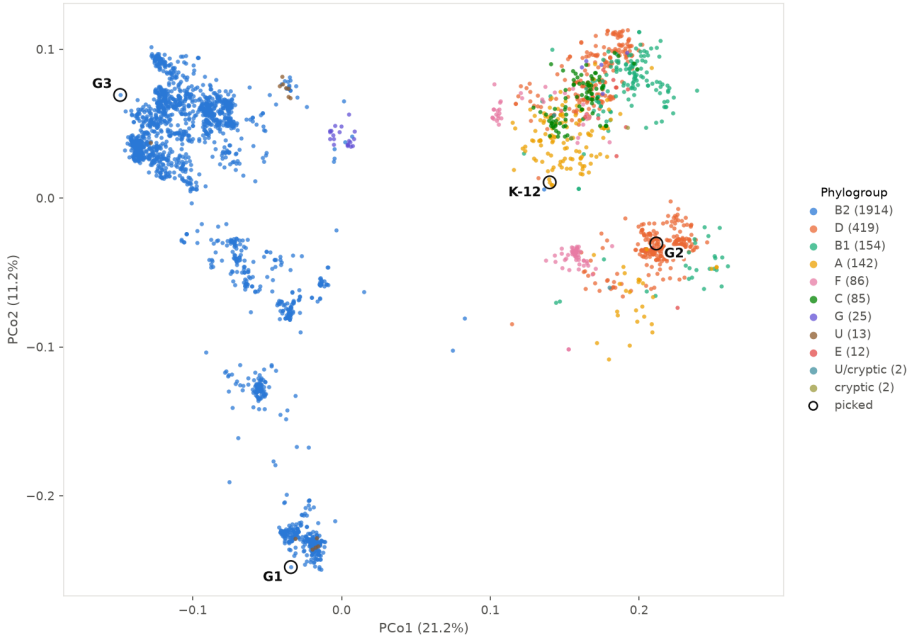


Figure 1: PCoA of gene content across N, W, U and M COG categories, coloured by phylogroup.

Label	Genome ID	Phylogroup	Size (Mb)	Genes	Set genes
K-12	K-12	A	4.64	4315	388
G1	2a94b780-2c0c-471e-b10d-87f35d3b8fa2	B2	5.44	5260	496
G2	abdbc0db-8027-42a1-ab67-7fd66d7574dd	D	5.60	5290	497
G3	dd652104-f063-4893-8fdc-1899ea8096dd	B2	5.91	5552	507

Table 1: Selected genomes based on Jaccard distances of genes in N, W, U and M COG categories. The table also shows the number of genes present in each genome, and the number of genes within the NWUM set.

2 Introduction

Most structural studies of *Escherichia coli* have used the laboratory strain K-12 or minicells derived from it. K-12 is not representative of the wider species. During decades of laboratory passage it has lost several hundred genes, including those needed to make the O-antigen and capsule, so the cell surface seen in these studies may differ substantially from that of clinical isolates. More recent work has examined individual structures and processes in detail, but how much cellular architecture varies across the *E. coli* population remains largely unknown. In this study, we aim to structurally characterise a range of clinically relevant *E. coli* strains to visualise this variability.

In order to select the strains to image using cryo-ET, we have analysed a set of 2,853 genomes from clinical isolates collected in Oxfordshire, UK (Lipworth et al. 2025). The aim of this initial analysis is to select three genomes which are the most dissimilar from one another (and from *E. coli* K-12), based on the presence and absence of genes within a set of structurally relevant functional categories.

3 Methods

3.1 Genome collection and quality control

We analysed 2,894 *Escherichia coli* genome assemblies from isolates collected in Oxfordshire, UK, between 2013 and 2018 and in 2020 as part of an observational study (Lipworth et al. 2025). Before further analysis, we filtered the assemblies on assembly quality and species identity.

We calculated assembly statistics with SeqKit v2.13.0 (`seqkit stats`). We excluded genomes with a total length outside 4.0–6.0 Mb, which can indicate an incomplete assembly or a different species, and genomes with an N50 below 50 kb. This removed 22 genomes, leaving 2,872.

We confirmed species identity with skani v0.4.0 (Shaw and Yu 2023), which estimates average nucleotide identity (ANI) and aligned fraction (AF) by approximate mapping rather than base-level alignment. We searched each assembly against the skani database for GTDB release 226 (skani_gtdb_r226-v0.3), which contains 743,475 bacterial and archaeal genomes (Parks et al. 2026). We discarded hits with an AF below 15% for either the query or the reference, and took the remaining hit with the highest ANI as each genome's top hit. We kept a genome only if its top hit was *E. coli* with an ANI of at least 95% and an AF of at least 60% for both the query and the reference. This removed a further 19 genomes, leaving a final set of 2,853. We also estimated completeness and contamination with CheckM2 v1.1.0 (Chklovski et al. 2023), but did not filter on them.

3.2 Characteristics of the final dataset

The 2,853 remaining genomes had a mean length of 5.13 Mb (SD 0.17 Mb; range 4.45–5.95 Mb), consistent with the expected genome size of *E. coli* (4.5–5.5 Mb), and a mean GC content of 50.58% (SD 0.12%). Assemblies had a median N50 of 212 kb (range 50–740 kb) and a median of 145 contigs (range 28–698). The median ANI to the closest *E. coli* genome in GTDB was 98.56% (range 96.32–99.92%), and at least 70% of each assembly aligned to its top hit. CheckM2 estimated a median

completeness of 100% (minimum 99.93%) and a median contamination of 0.71% (range 0–10.1%). Phylogroups had been assigned previously using ClermonTyping (Beghain et al. 2018; Lipworth et al. 2025). Most genomes belonged to phylogroup B2 (1,914; 67.1%), followed by D (419; 14.7%), B1 (154; 5.4%), A (141; 4.9%), F (86; 3.0%) and C (85; 3.0%). The remaining 54 genomes (1.9%) belonged to phylogroups G and E or were unassigned or cryptic.

3.3 Genome annotation and pangenome construction

We annotated 2,854 genomes with Prokka v1.15.6 (Seemann 2014): the 2,853 genomes that passed quality control and the *E. coli* K-12 MG1655 reference genome (RefSeq GCF_000005845.2). All genomes were annotated with the same parameters (`--genus Escherichia --species coli --compliant`). We built a pangenome from these annotations with Panaroo v1.8.0 (Tonkin-Hill et al. 2020), using moderate clean mode and removing invalid genes. Panaroo groups genes into clusters at a sequence identity threshold of 98%. It then builds a graph in which each gene cluster is a node and an edge joins two clusters that are adjacent in at least one genome. Panaroo uses this graph to correct annotation errors and to merge gene families. The pangenome contained 33,845 gene clusters of which, 3,075 were classified as core ($\geq$ 99% of genomes).

3.4 Functional annotation

We assigned functions to the gene clusters with eggNOG-mapper v3.0.0-beta6 (Cantalapiedra et al. 2021) and eggNOG database v7.0.0 (Hernández-Plaza et al. 2026), using DIAMOND searches (Buchfink et al. 2015). eggNOG-mapper was run on Panaroo's reference sequence for each cluster, which it translated using the bacterial genetic code (translation table 11). Of the 33,845 clusters, 26,218 (77.5%) had an eggNOG hit. Where eggNOG-mapper assigned a COG ID (14,190 clusters), we took its functional category from the NCBI COG 2020 definitions (Galperin et al. 2021). Where it gave only COG category letters without an ID (8,230 clusters), we used those letters. In total, 22,303 clusters (65.9%) were assigned at least one COG category (Table 2). A cluster can belong to more than one category.

3.5 Principal coordinate analysis of gene content

Principal coordinate analysis (PCoA) represents a matrix of pairwise distances in a small number of orthogonal axes. Each successive axis captures as much of the remaining variation between samples as possible. We used PCoA to compare the 2,854 genomes by the presence or absence of genes in selected functional categories.

For each set of COG categories, we included every gene cluster assigned to at least one category in the set. We then calculated pairwise Jaccard distances between genomes from the binary presence/absence matrix. Genomes with no genes in a set were excluded from that set's analysis. We calculated principal coordinates from each distance matrix by classical multidimensional scaling. We report the variance explained by each axis as its eigenvalue divided by the sum of the absolute values of all eigenvalues.

The main analysis used genes in four categories related to cell structure and the cell surface (N, W, U and M). We repeated the analysis with all genes, with combinations of these four categories, with individual categories, and with the three broad COG groups (Table 3).

3.6 Selection of maximally divergent genomes

We selected three genomes that, together with *E. coli* K-12, were as different from each other as possible in the main gene set (N, W, U and M). We used the Jaccard distances D calculated for the PCoA. With K-12 fixed, we chose the genomes a, b and c that maximised the smallest pairwise distance

COG category	Description
Information storage and processing	
J	Translation, ribosomal structure and biogenesis
A	RNA processing and modification
K	Transcription
L	Replication, recombination and repair
B	Chromatin structure and dynamics
Cellular processes and signalling	
D	Cell cycle control, cell division, chromosome partitioning
Y	Nuclear structure
V	Defense mechanisms
T	Signal transduction mechanisms
M	Cell wall/membrane/envelope biogenesis
N	Cell motility
Z	Cytoskeleton
W	Extracellular structures
U	Intracellular trafficking, secretion and vesicular transport
O	Posttranslational modification, protein turnover, chaperones
X	Mobilome: prophages, transposons
Metabolism	
C	Energy production and conversion
G	Carbohydrate transport and metabolism
E	Amino acid transport and metabolism
F	Nucleotide transport and metabolism
H	Coenzyme transport and metabolism
I	Lipid transport and metabolism
P	Inorganic ion transport and metabolism
Q	Secondary metabolites biosynthesis, transport and catabolism
Poorly characterised	
R	General function prediction only
S	Function unknown

Table 2: COG functional categories (NCBI COG 2020), grouped by broad function

among all four genomes:

$$\text{score}(a, b, c) = \min\big(D(a, b),\, D(a, c),\, D(b, c),\, D(a, \text{K12}),\, D(b, \text{K12}),\, D(c, \text{K12})\big)$$

Rather than scoring every possible trio, we used a search that is guaranteed to find the best one. First, we built K-12 into the pairwise distances. For each pair of candidate genomes, we defined an adjusted distance

$$D'(x, y) = \min\big(D(x, y),\, D(x, \text{K12}),\, D(y, \text{K12})\big),$$

the smallest distance among x, y and K-12. For example, two genomes that are far apart from each other but where one is close to K-12 have a small adjusted distance. Because each adjusted distance already includes both genomes' distances to K-12, the smallest of a trio's three adjusted distances equals its score, so the problem reduces to choosing three genomes with K-12 no longer needed. Next, we went through pairs of genomes from the largest to the smallest adjusted distance. For each pair, we found the third genome giving the highest score. A trio's score can never exceed the adjusted distance of any pair within it, so once the next pair's adjusted distance was no larger than the best score found so far, no remaining trio could do better and we stopped.

Gene set	COG categories	Gene clusters
All genes	all	33,845
Information storage and processing	J, A, K, L, B	2,976
Cellular processes and signalling	D, Y, V, T, M, N, Z, W, U, O	4,899
Metabolism	C, G, E, F, H, I, P, Q	3,282
Motility, extracellular structures, secretion and envelope (main analysis)	N, W, U, M	2,748
Motility, extracellular structures, secretion, envelope and defence	N, W, U, M, V	3,586
Motility, extracellular structures, envelope and defence	N, W, M, V	3,052
Motility, extracellular structures and secretion	N, W, U	1,254
Translation and ribosomal structure	J	475
Transcription	K	1,220
Replication, recombination and repair	L	1,340
Defence	V	854
Envelope	M	1,558
Energy production	C	461
Inorganic ion transport	P	460

Table 3: Gene sets used for Jaccard distances and PCoA. A gene cluster is in a set if it is assigned to at least one of the set's COG categories. All 2,854 genomes had genes in every set.

3.7 Antimicrobial resistance genes

For the selected clinical genomes, we took antimicrobial resistance (AMR) genes and point mutations from the AMRFinderPlus v3 (Feldgarden et al. 2021) results of Lipworth et al. (2025). We kept only AMR elements, excluding stress response and virulence genes and the efflux class. We kept hits that were exact or allele matches to a reference (EXACTX, ALLELEX), full-length matches with at least 90% identity (BLASTX), or known resistance point mutations (POINTX, POINTN). We excluded partial hits, hits with an internal stop codon and HMM-only hits. Where a hit's subclass listed several drugs, we counted it under each drug.

3.8 Code availability

The analysis pipeline (Nextflow DSL2) and analysis scripts are available at `https://github.com/igoodridge/e-coli-pangenome`. Software versions are pinned in the conda environment files in `envs/`, and QC thresholds, COG gene sets and genome selection settings are in `conf/`.

4 Results

4.1 Pangenome and functional annotation

The pangenome of the 2,854 genomes contained 33,845 gene clusters, most of which were rare. 27,516 clusters (81.3%) were cloud genes, found in fewer than 15% of genomes, while 3,075 (9.1%) were core, 211 (0.6%) soft core and 3,043 (9.0%) shell.

22,303 clusters (65.9%) were assigned at least one COG category (Figure 2). The largest category was S (function unknown; 8,781 clusters), followed by X (mobilome; 2,255), M (cell wall, membrane and envelope biogenesis; 1,558), L (replication, recombination and repair; 1,340) and K (transcription;

1,220). Categories for essential cellular functions had the largest share of core clusters: nearly half of the clusters in translation (J), energy production (C), amino acid metabolism (E) and nucleotide metabolism (F) were present in at least 95% of genomes. By contrast, most clusters in the four categories used for genome selection were accessory, with only 11% of N, 5% of W, 5% of U and 13% of M clusters being core. Almost all mobilome clusters (2,252 of 2,255) were accessory.

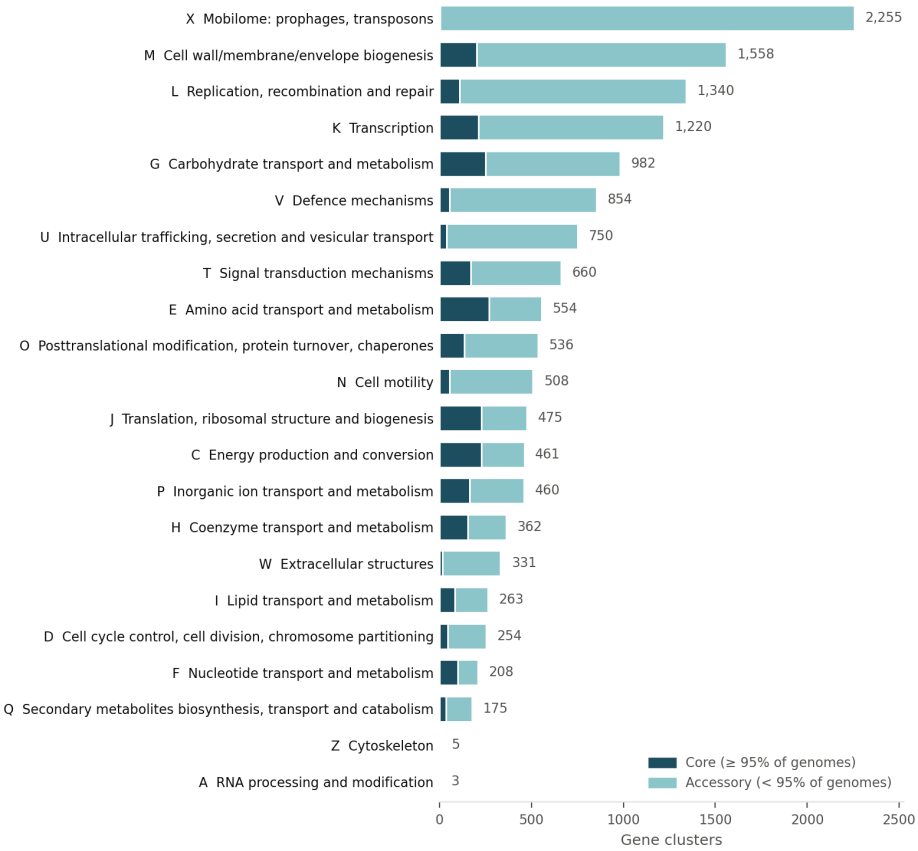


Figure 2: Number of gene clusters in each COG category, split into core (present in at least 95% of genomes) and accessory clusters. A cluster assigned to several categories is counted in each. The poorly characterised categories R (general function prediction only) and S (function unknown) are not shown, nor are B and Y which correspond to chromatin structure and nuclear structure respectively, which had no clusters, as expected.

4.2 Selected genomes

With K-12 fixed, the three genomes selected using the N, W, U and M gene set were G1 and G3 from phylogroup B2 and G2 from phylogroup D (Table 1). The smallest pairwise Jaccard distance among the four genomes was 0.419, between K-12 and G1 (Table 4). This is greater than 91.2% of all pairwise distances between genomes in this gene set (median 0.335; maximum 0.513).

The selected genomes are 5.44–5.91 Mb long and carry 5,260–5,552 genes, compared with 4.64 Mb and 4,315 genes in K-12. Each has 496–507 genes in the N, W, U and M set, compared with 388 in K-12.

Although the selection used only the N, W, U and M genes, the four genomes are also well separated

by their whole gene content: their pairwise distances across all genes range from 0.365 (K-12 and G2) to 0.452 (G2 and G3), and the smallest of these is greater than 66.5% of all pairwise distances (Table 4).

Table 1: Selected genomes

Label	Genome ID	Phylogroup	Size (Mb)	Genes	Set genes
K-12	K-12	A	4.64	4315	388
G1	2a94b780-2c0c-471e-b10d-87f35d3b8fa2	B2	5.44	5260	496
G2	abdbc0db-8027-42a1-ab67-7fd66d7574dd	D	5.60	5290	497
G3	dd652104-f063-4893-8fdc-1899ea8096dd	B2	5.91	5552	507

	K-12	G1	G2	G3
K-12	–	0.419	0.431	0.441
G1	0.396	–	0.444	0.433
G2	0.365	0.424	–	0.472
G3	0.428	0.390	0.452	–

Table 4: Pairwise Jaccard distances between the selected genomes. Upper triangle: N, W, U and M gene set, used for the selection. Lower triangle: all genes.

Notably, all three selected genomes carry multiple antimicrobial resistance (AMR) genes, in contrast with *E. coli* K-12, which carries none (Table 5).

Table 5: AMR genes and point mutations per selected genome, grouped by drug

Drug	G1	G2	G3
Ampicillin			marR_S3N
Beta-lactam	blaEC	blaEC, blaTEM-1	blaEC
Cephalosporin		blaCTX-M-14	
Chloramphenicol			marR_S3N
Colistin	pmrB_E123D		pmrB_E123D
Fosfomycin	glpT_E448K, ptsI_V25I, uhpT_E350Q	glpT_E448K	glpT_E448K
Fosmidomycin		cyaA_S352T	
Gentamicin		aac(3)-IId	
Quinolone	gyrA_D87N, gyrA_S83L, parC_E84V, parC_S80I, parE_I529L		marR_S3N
Rifampin			marR_S3N
Streptomycin		aadA1	
Streptothricin		sat2	
Tetracycline	tet(A)		marR_S3N
Trimethoprim		dfrA1	

4.3 PCoA of cell surface and secretion genes

In the N, W, U and M gene set, the first two principal coordinates explained 21.2% and 11.2% of the variation (Figure 1; Figure S1). PCo1 separated phylogroup B2 from the other phylogroups, with phylogroup G between them. PCo2 split the B2 genomes into several distinct clusters, and the D genomes into two. The selected genomes lie at the edges of this distribution: G1 and G3 at opposite ends of the B2 genomes along PCo2, G2 in the lower of the two D clusters, and K-12 at the edge of the cluster containing phylogroups A, B1, C and D.

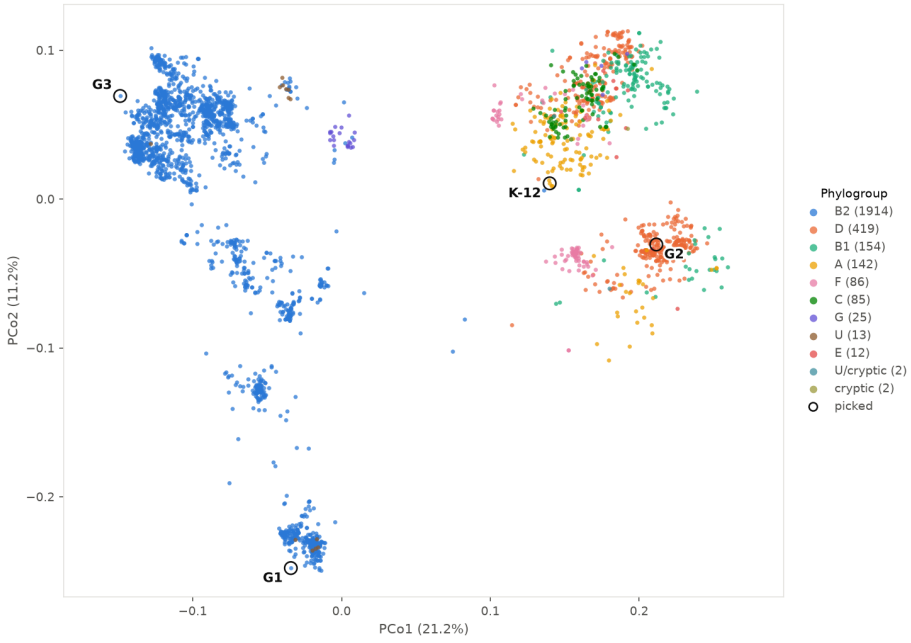


Figure 1: PCoA of gene content across N, W, U and M COG categories, coloured by phylogroup.

4.4 PCoA of all genes

We repeated the PCoA with each of the other gene sets in Table 3. In every gene set, PCo1 separated most B2 genomes from the other phylogroups, although this separation was weakest for replication, recombination and repair genes (L; Figure 12).

Across all genes, PCo1 and PCo2 explained 20.3% and 8.1% of the variation (Figure 3). As in the N, W, U and M gene set, G1 and G3 lie at opposite ends of the B2 genomes, but G2 lies much closer to K-12 than to either B2 genome. This matches their pairwise distances, as K-12 and G2 are the closest pair across all genes (Table 4).

4.5 PCoA of other combinations of cell surface categories

Changing the combination of cell surface categories had little effect on the overall structure (Figures 7 to 9). Adding defence genes (V) to the main gene set gave almost the same plot (PCo1 21.3%, PCo2 10.4%). Removing secretion genes (U) from this set (PCo1 19.4%, PCo2 11.6%), or leaving out envelope genes (M) and defence genes (PCo1 23.8%, PCo2 12.1%), split the D genomes into more distinct clusters. In all three sets, the four selected genomes stayed at the edges of the distribution, with G1 and G3 at opposite ends of the B2 genomes and G2 in a separate cluster of D genomes.

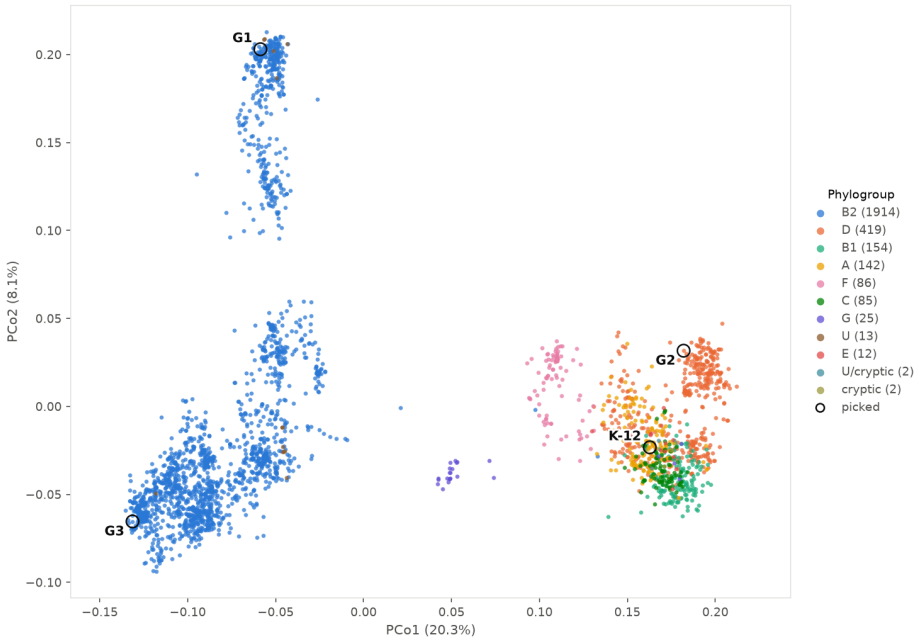


Figure 3: PCoA of gene content across all genes, coloured by phylogroup.

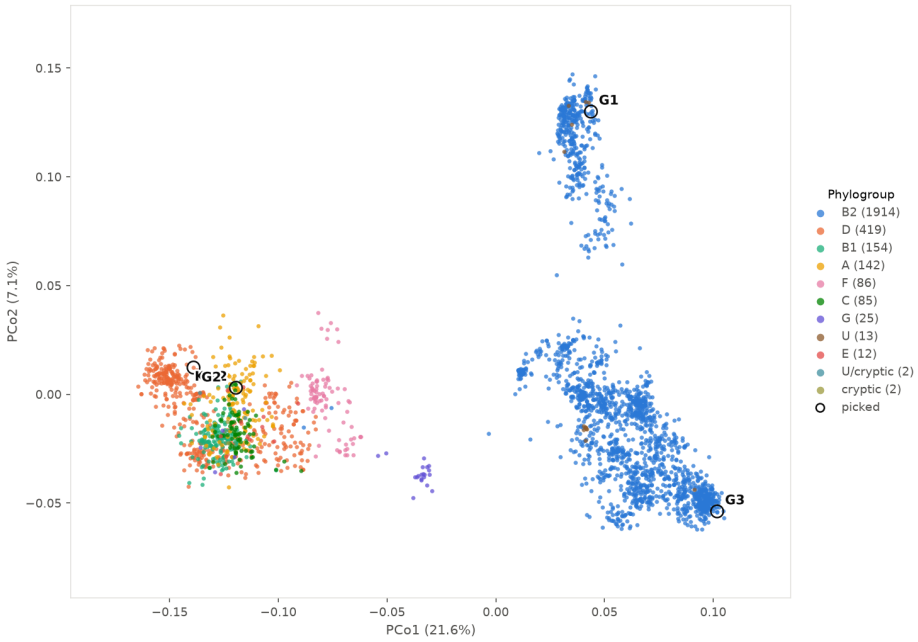


Figure 4: PCoA of gene content across information storage and processing genes (COG categories J, A, K, L and B), coloured by phylogroup.

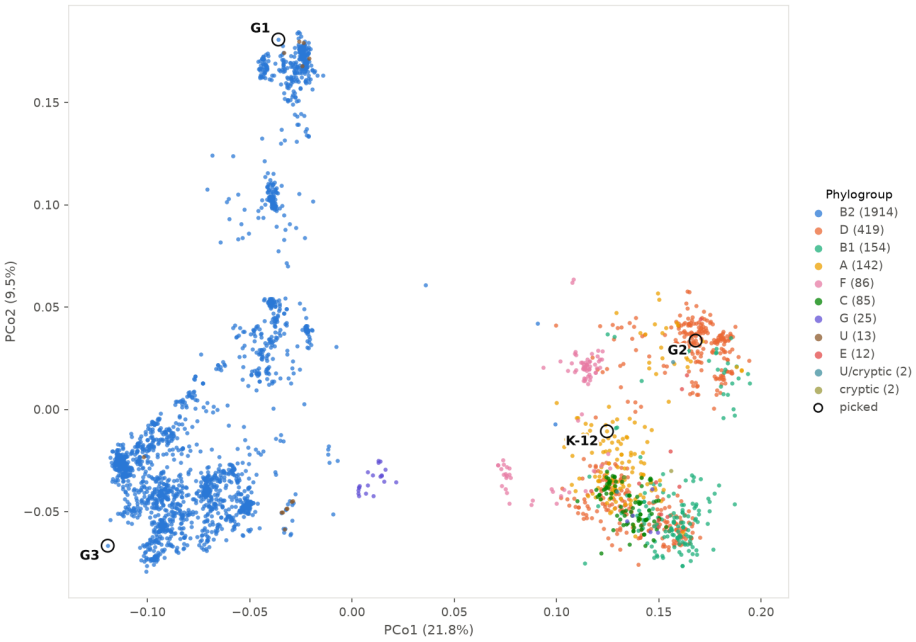


Figure 5: PCoA of gene content across cellular processes and signalling genes (COG categories D, Y, V, T, M, N, Z, W, U and O), coloured by phylogroup.

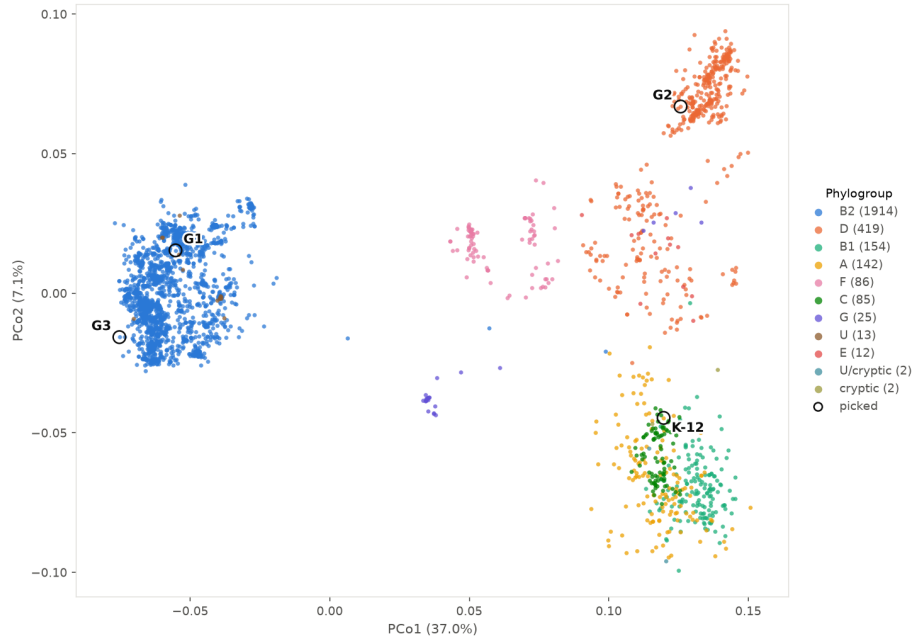


Figure 6: PCoA of gene content across metabolism genes (COG categories C, G, E, F, H, I, P and Q), coloured by phylogroup.

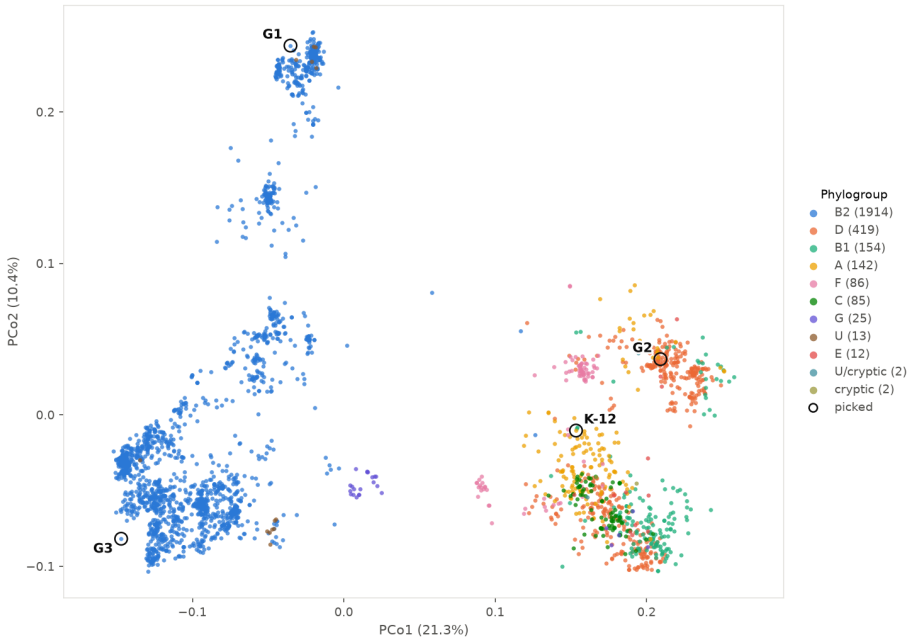


Figure 7: PCoA of gene content across motility, extracellular structures, secretion, envelope and defence genes (COG categories N, W, U, M and V), coloured by phylogroup.

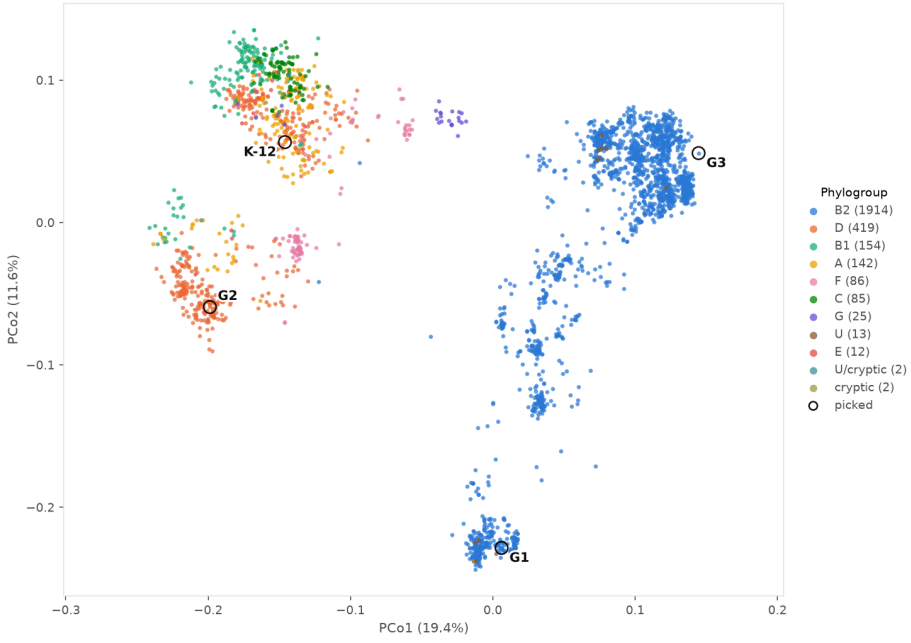


Figure 8: PCoA of gene content across motility, extracellular structures, envelope and defence genes (COG categories N, W, M and V), coloured by phylogroup.

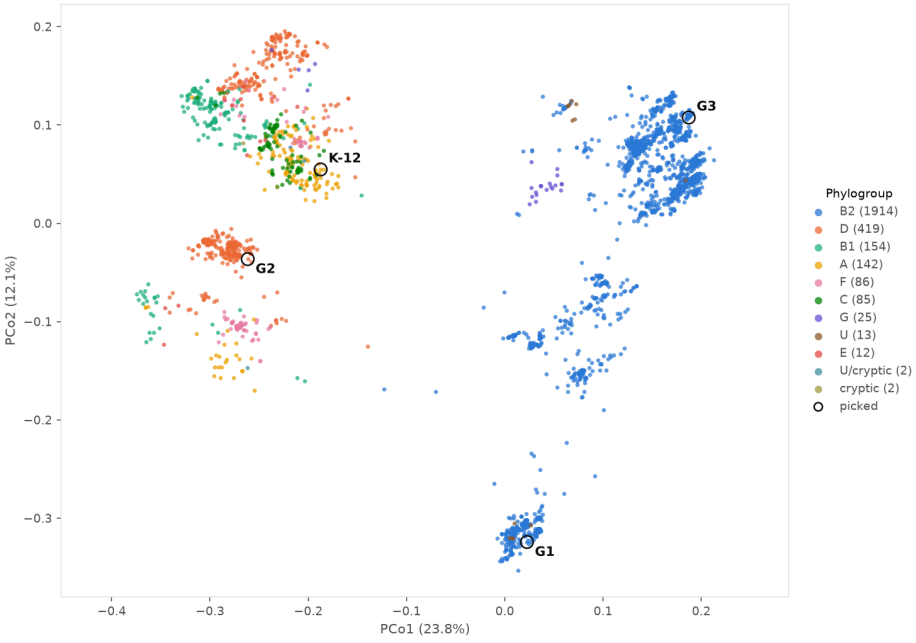


Figure 9: PCoA of gene content across motility, extracellular structures and secretion genes (COG categories N, W and U), coloured by phylogroup.

4.6 PCoA of single COG categories

The variation explained by PCo1 differed widely between single categories (Figures 10 to 16). It was highest for energy production and conversion genes (C; 44.3%), the highest of any gene set, followed by inorganic ion transport (P; 31.6%) and transcription (K; 30.6%). It was lowest for replication, recombination and repair genes (L; 10.4%), where the B2 genomes formed a gradient rather than a separate cluster.

The selected genomes were not always well separated in single categories. In energy production and inorganic ion transport genes, G1 and G3 lie close together within the B2 genomes, and in transcription genes K-12 and G2 lie close together. In defence genes (V), G1 and G3 lie at opposite ends of the B2 genomes, and in envelope genes (M), G3 lies at the outer edge of the B2 genomes.

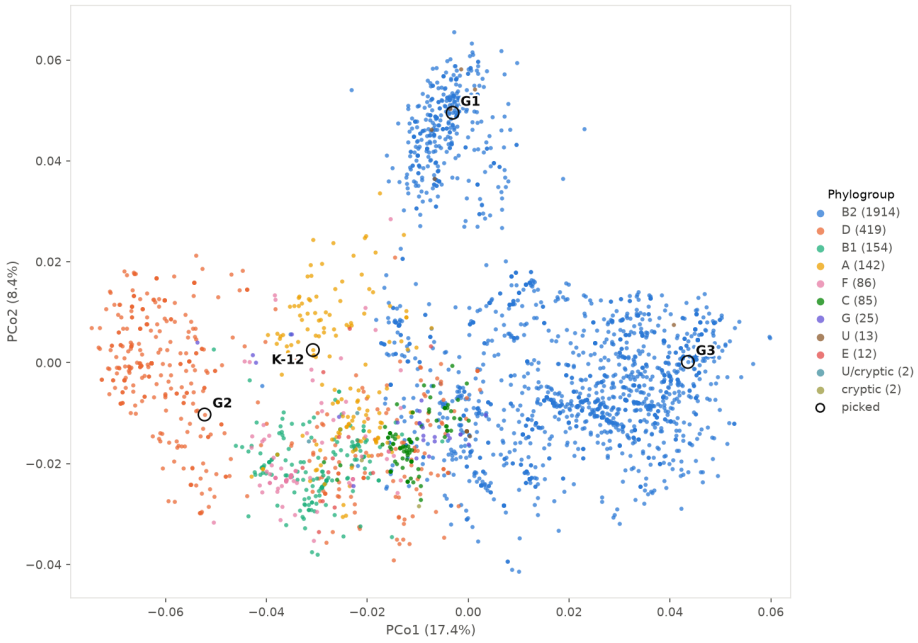


Figure 10: PCoA of gene content across translation and ribosomal structure genes (COG category J), coloured by phylogroup.

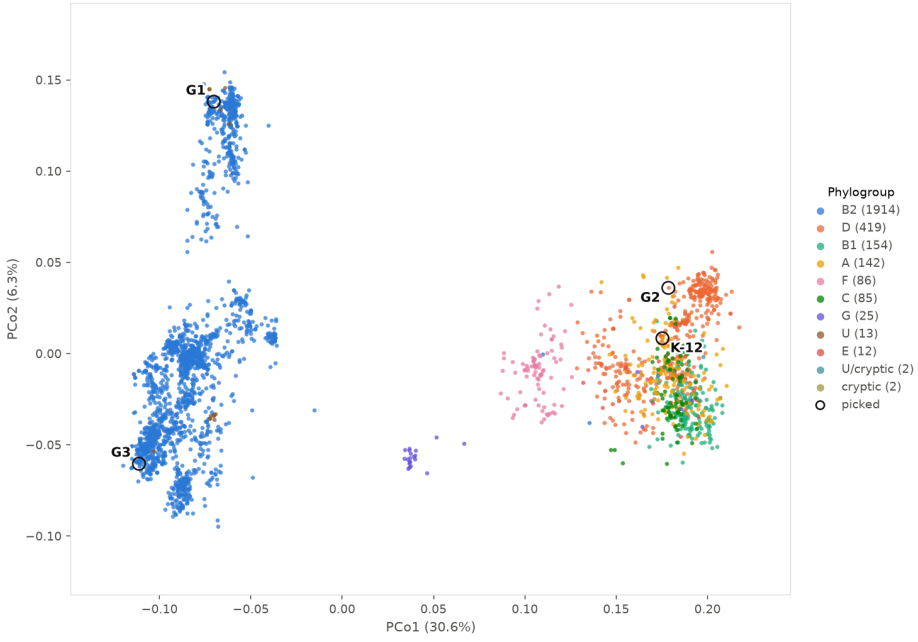


Figure 11: PCoA of gene content across transcription genes (COG category K), coloured by phylogroup.

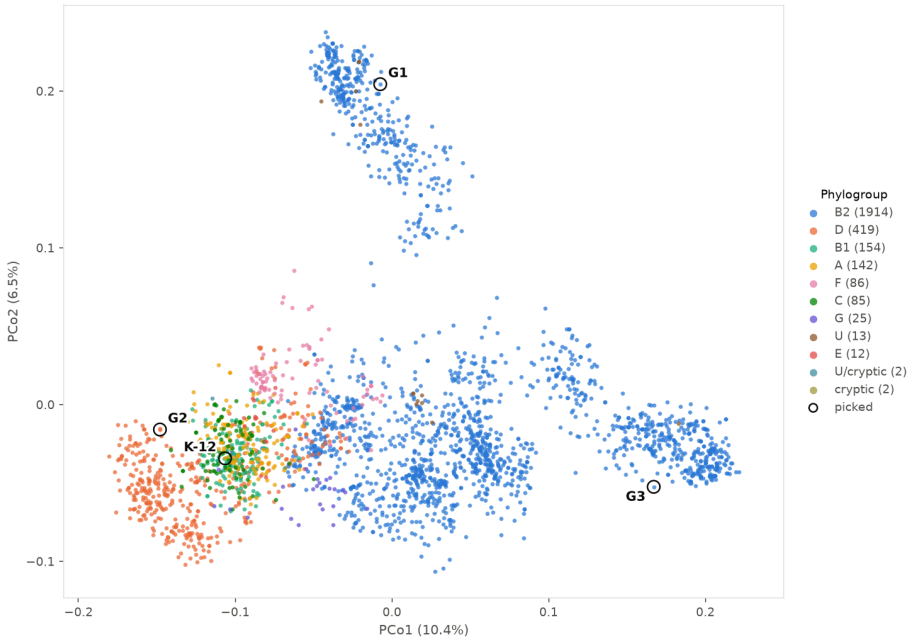


Figure 12: PCoA of gene content across replication, recombination and repair genes (COG category L), coloured by phylogroup.

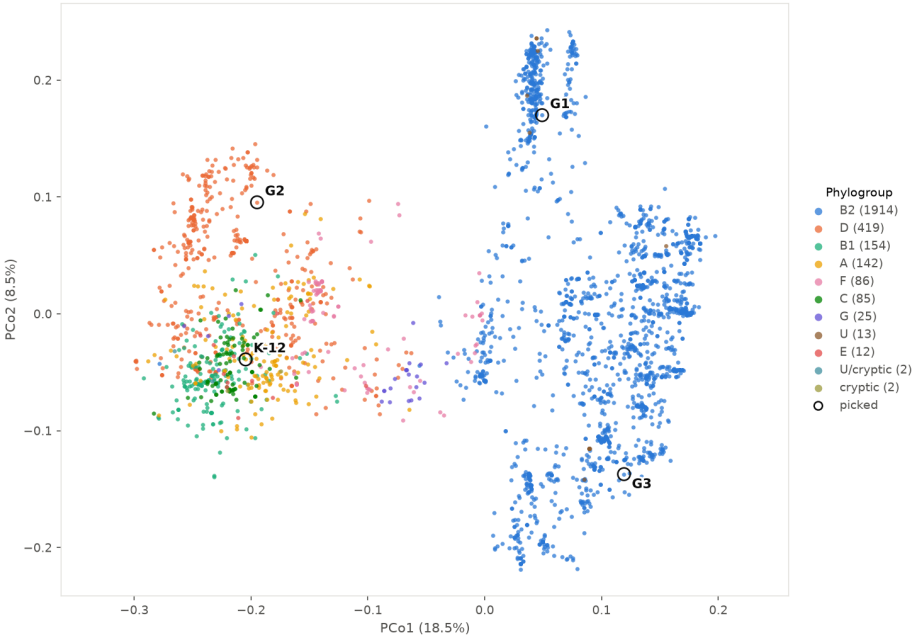


Figure 13: PCoA of gene content across defence genes (COG category V), coloured by phylogroup.

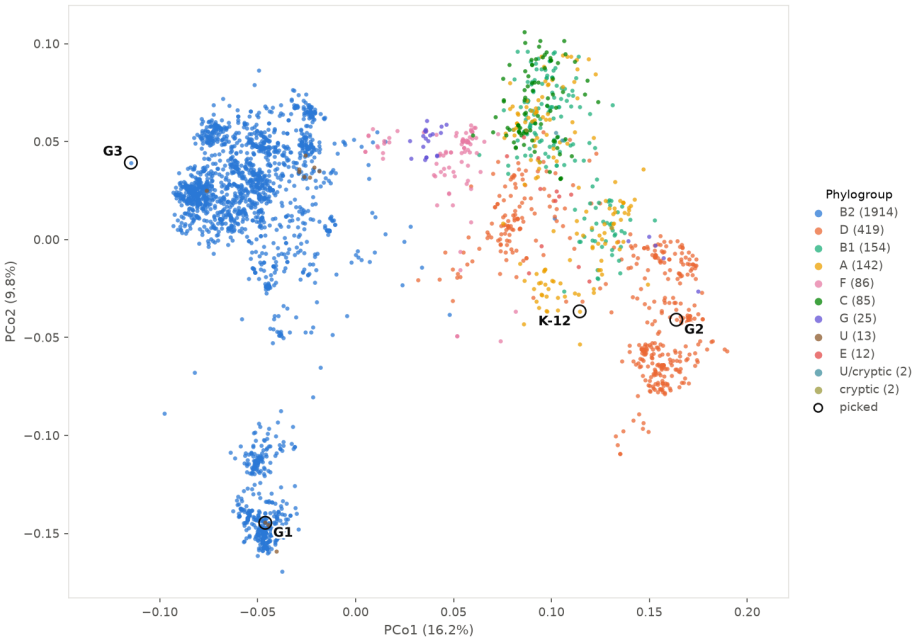


Figure 14: PCoA of gene content across cell wall, membrane and envelope biogenesis genes (COG category M), coloured by phylogroup.

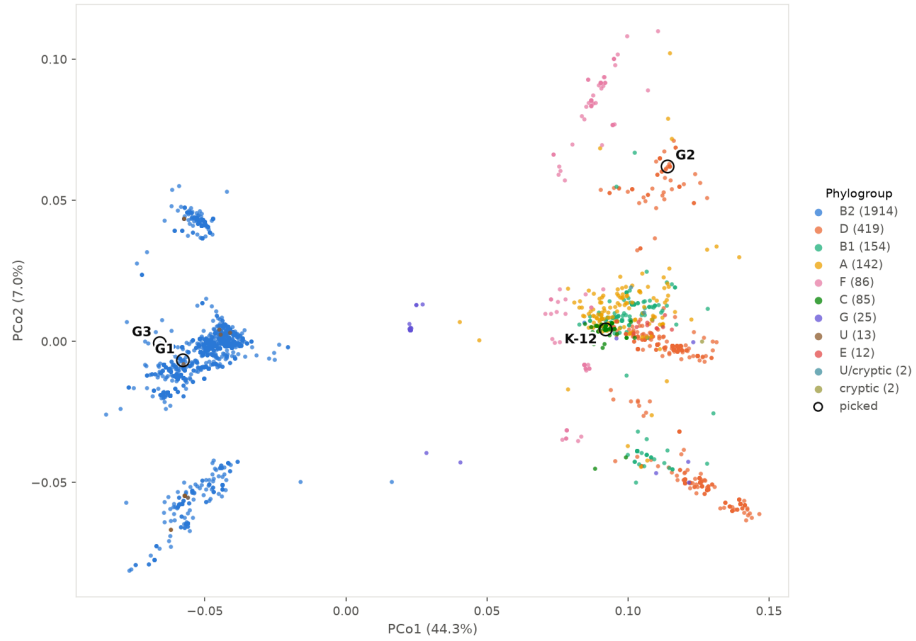


Figure 15: PCoA of gene content across energy production and conversion genes (COG category C), coloured by phylogroup.



Figure 16: PCoA of gene content across inorganic ion transport and metabolism genes (COG category P), coloured by phylogroup.

5 Further considerations

5.1 Gene categories

We could consider the inclusion of other gene categories, for example defence, within the set of genes we have selected.

5.2 Genome selection

We could consider alternative methods for selecting the genomes that are most different from one another. Furthermore, we could consider requiring that each selected genome originate from a different phylogroup.

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6 Supplementary

6.1 Variance explained by each principal coordinate

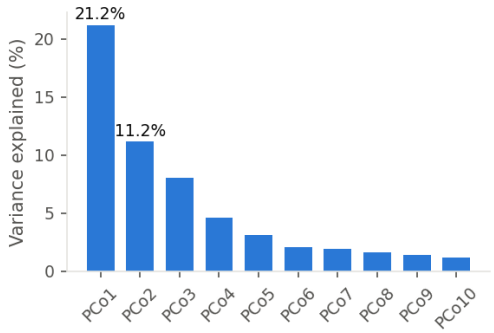


Figure S1: Variance explained by the first ten principal coordinates in the PCoA of the N, W, U and M gene set.

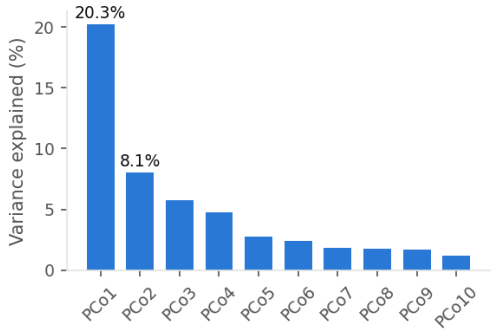


Figure S2: Variance explained by the first ten principal coordinates in the PCoA of all genes.

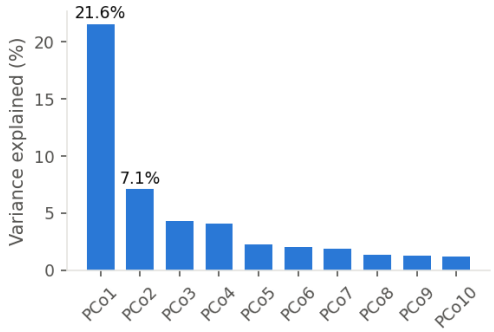


Figure S3: Variance explained by the first ten principal coordinates in the PCoA of information storage and processing genes (COG categories J, A, K, L and B).

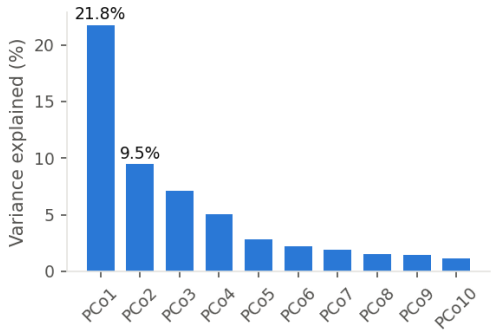


Figure S4: Variance explained by the first ten principal coordinates in the PCoA of cellular processes and signalling genes (COG categories D, Y, V, T, M, N, Z, W, U and O).

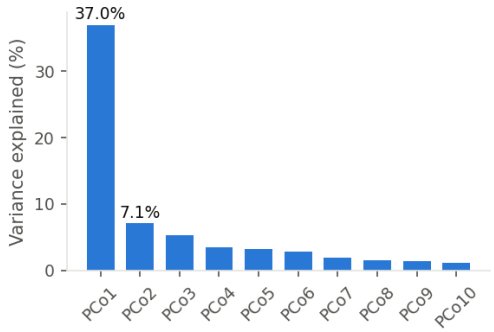


Figure S5: Variance explained by the first ten principal coordinates in the PCoA of metabolism genes (COG categories C, G, E, F, H, I, P and Q).

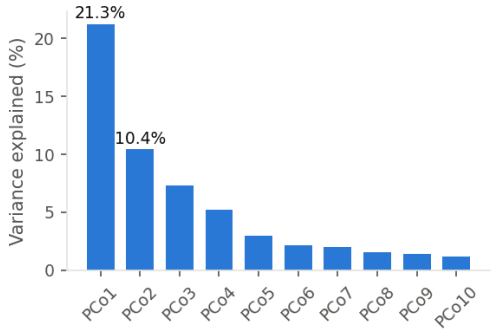


Figure S6: Variance explained by the first ten principal coordinates in the PCoA of motility, extracellular structures, secretion, envelope and defence genes (COG categories N, W, U, M and V).

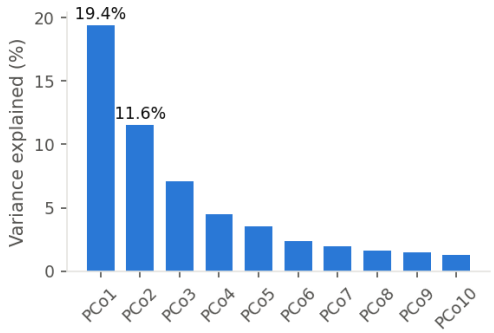


Figure S7: Variance explained by the first ten principal coordinates in the PCoA of motility, extracellular structures, envelope and defence genes (COG categories N, W, M and V).

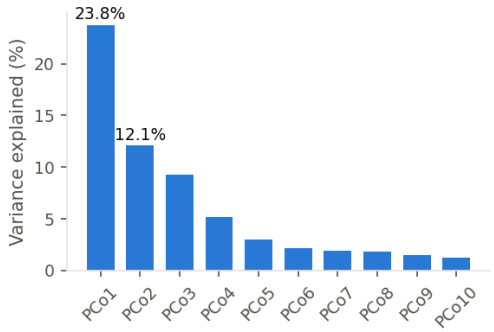


Figure S8: Variance explained by the first ten principal coordinates in the PCoA of motility, extracellular structures and secretion genes (COG categories N, W and U).

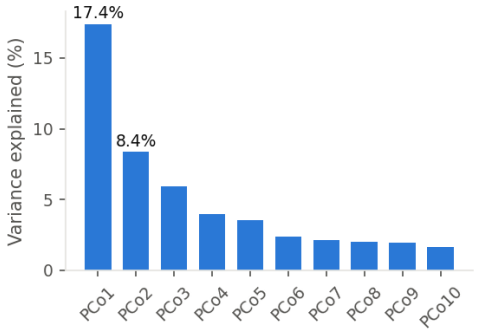


Figure S9: Variance explained by the first ten principal coordinates in the PCoA of translation and ribosomal structure genes (COG category J).

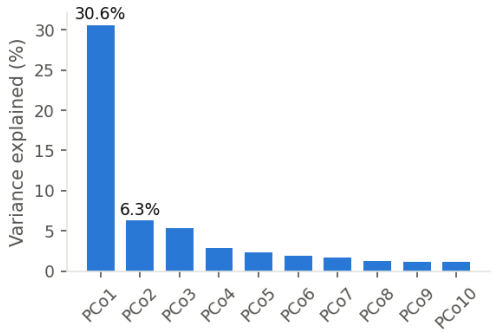


Figure S10: Variance explained by the first ten principal coordinates in the PCoA of transcription genes (COG category K).

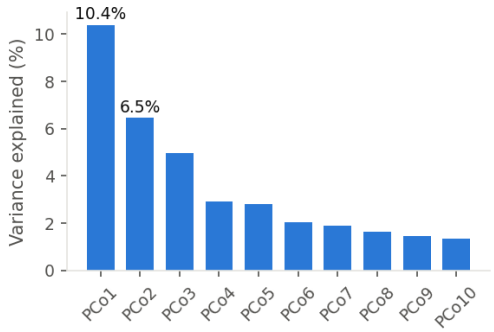


Figure S11: Variance explained by the first ten principal coordinates in the PCoA of replication, recombination and repair genes (COG category L).

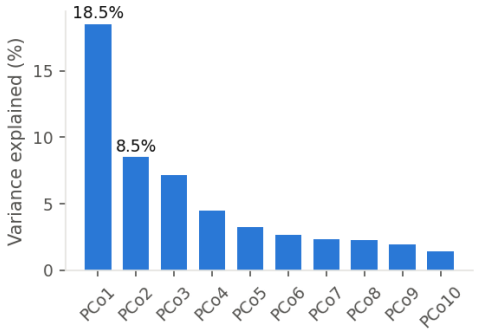


Figure S12: Variance explained by the first ten principal coordinates in the PCoA of defence genes (COG category V).

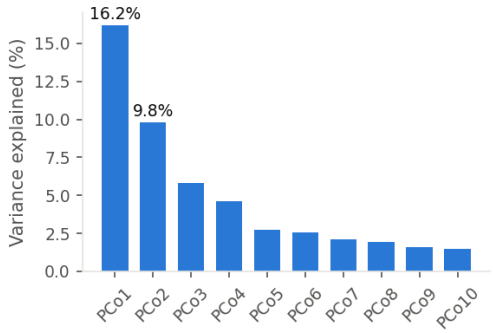


Figure S13: Variance explained by the first ten principal coordinates in the PCoA of cell wall, membrane and envelope biogenesis genes (COG category M).

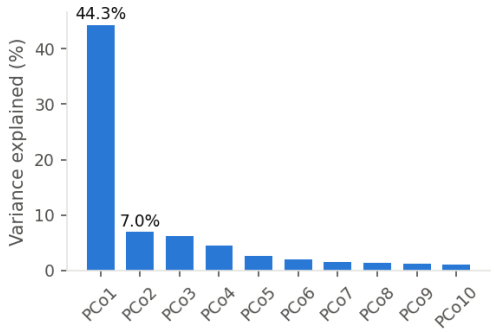


Figure S14: Variance explained by the first ten principal coordinates in the PCoA of energy production and conversion genes (COG category C).

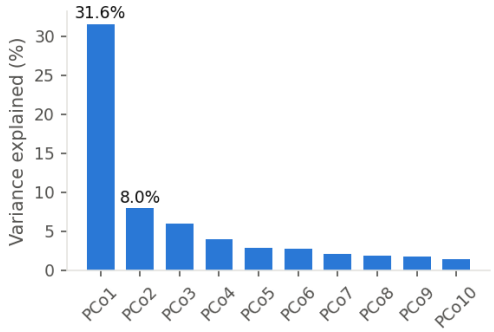


Figure S15: Variance explained by the first ten principal coordinates in the PCoA of inorganic ion transport and metabolism genes (COG category P).